Part 1: SQL.txt, ER Design

Client Table Considerations:

- clientID PK INT
- firstName, lastName VARCHAR
- address FK
- phoneNumber, email VARCHAR
- creditCardNumber INT, CVV INT, Expiration DATE

- **Relationships**:
    - Client:
        - A client may submit a request for a flooring project
    - Contractor:
        -

Address Table:

- addressID PK
    - CONSTRAINT UNIQUE_ADDR UNIQUE (address, postalCode, city, state, country)
- address, postalCode, city, state, region VARCHAR
- country CHAR

Flooring Product Table Considerations:

- productID
- name, flooringType, manufacturer, ~~material~~, color VARCHAR
- unit price FLOAT; pricing unit VARCHAR
- product dimensions (l*w) FLOAT
    - Constraint, all must exist
    - If not, then it is nullable possibly
- square foot coverage FLOAT
    - Nullable or empty possible
- description TEXT
    - nullable or empty possible

- **Relationships**:
    - Client:
        - Client chooses products when having to choose a flooringProjectArea
    - Contractor:
        - The exact flooring products and prices are maintained by the contractor through the website.

Flooring Project Table

- clientID, flooringProjectID (can just be clientID & serviceAddressID combo)
- Service Address

    - Can refer to an Address Table
- proposedBudget FLOAT
- optionalNotes TEXT
- contractorAccepted BOOLEAN
    - true (by contractor; which indicates the quote phase)
    - false (no quotes generated)
- Photos
    - Clients may upload 5 associated photos; blob? Better to have directories

- ~~Room/Area Name VARCHAR~~
- ~~Total Square Footage FLOAT~~
    - ~~Not inputtable but rather system calculated~~
- ~~Desired flooring material/product~~
    - ~~Can have a dropdown unique selections for materials~~
    - ~~Set of flooring products (this could potentially be its own table, helps with the flooring project and quote table)~~
- ~~Current Flooring Type~~
    - ~~If applicable, nullable otherwise~~
- ~~Whether removal of existing floor is required BOOLEAN~~
- ~~Set of installation preferences~~
- ~~Preferred Installation date or date range (startDate, endDate)~~
    - ~~endDate nullable, if present, then it represents a range~~
    - ~~startDate NOT nullable~~

- **Relationships**:
    - A project may contain one or more rooms or flooring areas.
    - The system should calculate or store the total requested floor area based on the rooms included in the request.
    - Client:
        - For each room or area, the client specifies the attributes
    - Contractor:
        - Contractor can reject the request, with an explanation

Flooring Project Area Table
- flooringProjectID FK INT
- Room/Area Name VARCHAR
- Total Square Footage FLOAT
    - Not inputtable but rather system calculated
- Desired flooring material & product (Project Area Products)
    -
- Current Flooring Type
    - If applicable, nullable otherwise
- Whether removal of existing floor is required BOOLEAN
- Installation Preferences TEXT

- Preferred Installation date or date range (startDate, endDate)
    - endDate nullable, if present, then it represents a range
    - startDate NOT nullable
- isUpgrade BOOLEAN
    - Relevant for billing! DEFAULT FALSE

- **Relationships**:
    - Dropdown unique selection for materials using the MaterialOptions Table

Project Area Products Table:
- projectAreaProductID PK
    - flooringProjectAreaID FK
- flooringProductID FK - each row to allow storing of sets of products per area
    - Can have a dropdown unique selections for materials; materialID FK, nullable
    - Set of flooring products, stored as rows in this table; flooringProductID FK
- allocatedSquareFootage
    - Constraint must be that the combination of the square footage must not be greater than total square footage of the flooring project area FK

MaterialOptions Table:
- materialID
- name VARCHAR
- description TEXT
- unitPrice FLOAT
- pricingUnit
    - ENUM: squareFoot

ProductMaterialMapping Table:
- productID, materialID FK
    - Different products may have differing options of materials to choose from
    - Note that the flooring product table does not have materialID FK, but rather, the client chooses
- Relationships:
    - Client chooses material option in the flooring project area

Quote Table (separate line items)

- quoteID
- flooringProjectID
    - Material as noted in the Flooring Project
        - Set of Flooring Material; handled by Flooring Project Area Table
        - Set of Quantity of Flooring Material; handled by Flooring Project Area Table
            - Can be calculated for certain
        - Set of Material Cost; handled by Flooring Project Area Table
    - Removal of existing flooring FLOAT; handled
- Quote Date/Time DATETIME
    - This can determine order given a client ID
- TotalPrice FLOAT
- Proposed Installation Date or date range (startDate, endDate) DATE
    - Project WIDE, not per product
- note VARCHAR
    - Can be nullable or empty
- Waste Percentage FLOAT
    - Calculated from the total of flooring material quantity, multiplied by the percentage to get waste cost
- Installation Labor, Underlayment, Trim/Baseboard work, stairs, delivery FLOAT
- Other Charges or Discounts
    - One set for charges FLOAT; corresponding set for description
    - One set for discounts FLOAT; corresponding set for description
- quoteStatus INT
    - < 0 means quote is renegotiating
    - = 0 means quote is rejected
    - > 0 means quote is accepted

- **Relationships**:
    - Every quote, counterproposal, and response must remain stored in the database. Previous versions must not be overwritten because they may be needed later if a disagreement occurs.
    - Client:
        - May accept quote, creating installation order
        - Reject/Cancel quote
        - Renegotiate quote (counterproposal or note)
            - flooringProductPrice should not be disputable, but other charges should be
    - Contractor:
        - May respond to client with another quote

Order Table

- orderID, quoteID, flooringProjectID (redundant)
    - clientID is stored in quoteID via flooringProjectID
    - quoteID contains flooring products,

    - rooms/areas are included from the quote's flooring project ID; furthermore, the quote has the agreed price, as well as scheduled installation
- actualMaterialQuantity FLOAT
- actualInstallationDate DATE, actualProjectCompletionDate DATE
- completionNotes TEXT
    - Optional
- orderStatus VARCHAR
    - scheduled, in progress, completed, cancelled

- **Relationships**:
    -

Billing Table
- billingID, orderID
- charges, discounts, credits
    - Set of values in FLOAT
    - Set of descriptions in TEXT
- additionalFlooringMaterial FLOAT
- additionalLabor FLOAT
- customerUpgrades
    - Set of upgrade descriptions TEXT
    - Corresponding Set of upgrade costs FLOAT

- **Relationships**:
    - DIsputes continue until the bill is paid or remains under dispute
    - Client:
        - May pay the bill using the registered credit card, OR dispute the bill and enter a note explaining the reason
    - Contractor:
        - May respond to disputes by explaining the charge, adjusting the bill, issuing a discount or credit, or generating a revised bill

CustomLineItems Table: To hold additional charges, discounts, credits
- customLineItemID
- quoteID AND billingID (in the same vein as the disputes table)
- chargeType
    - ENUM: Charge, Discount, Credit
- description TEXT
- quantity
    - NULLABLE indicates flat value (x1)
- unitPrice, totalPrice FLOAT

Disputes Table: To negotiate costs, stores responses
- quoteID AND billingID; or just have flooringProjectID

    - clientID is stored in quoteID via flooringProjectID; we can store it redundantly here
    - IF Having both ID types be FKs and be nullable
        - CONSTRAINT chk_dispute_exclusive_ref CHECK (
            (quote_id IS NOT NULL AND billing_id IS NULL) OR
            (quote_id IS NULL AND billing_id IS NOT NULL)
)
    - flooringProjectID can be found directly in the quote table, or through orderID's stored quoteID
    - Might be best to have flooringProjectID, and quoteID and billingID in this disputes table
        - This is because a flooringProject may not have an associated quoteID
- response TEXT
- isClientDispute BOOLEAN

- **Relationships**:
    - Deals with the quote and order tables primarily